

For these exercises, you should read the notes on [Recurrent Neural Networks](#).

1) Representing sequences

We are interested in making a neural network that can predict the next element in a sequence. For each of the sequences below, indicate *all* of the following models that could make correct predictions if its weights were set correctly (don't worry about learning). The possible models are:

- A **bigram** model, which is a feedforward network that takes sequence element y_{t-1} as input and generates element y_t as output.
- A **trigram** model, which is a feedforward network that takes sequence elements y_{t-2} and y_{t-1} as input and generates element y_t as output.
- A simple **recurrent** model, with a one-dimensional state, governed by the equations: $s_t = \tanh(W^{s,s}s_{t-1} + W^{s,x}y_{t-1})$ and $P(y_t = k \mid y_0, \dots, y_{t-1}) = \text{softmax}(W^o s_t)$.

1A)

1, 1, -1, 1, 1, -1, 1, 1, -1, ...

- ☐ bigram
- ☒ trigram
- ☒ recurrent

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1B)

1, -1, 1, -1, 1, -1, ...

- ☒ bigram
- ☒ trigram
- ☒ recurrent

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1C)

1, 1, 1, -1, 1, 1, 1, -1, ...

- ☐ bigram
- ☐ trigram
- ☒ recurrent

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2) RNN Learning

Consider a very simple RNN, defined by the following equations:

$$s_t = ws_{t-1} + x_t$$
$$p_t = s_t$$

There is a single scalar parameter w . Assume that $s_0 = 0$. Note that p_t is the actual output of the RNN at time t , and we want this to equal the corresponding true label y_t .

2A) For an RNN that, given $x = [1, 1, 1]$ produces $p = [1, 2, 3]$, what is the value of w ?

Enter a number: 1

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2B) Assume that the loss at a single time step is $\text{Loss}(p_t, y_t) = \frac{1}{2}(p_t - y_t)^2$. Recall p_t is the actual output and y_t is the target output. What is $\frac{d\text{Loss}}{dw}$ where $\text{Loss} = \sum_{t=1}^3 \text{Loss}(p_t, y_t)$? Assume $s_0 = 0$.

Enter an expression in terms of (only) w , x_1 , x_2 , x_3 , y_1 , y_2 , y_3 . Do not use any of the p_t .

$$d\text{Loss}/dw = (x_1*w + x_2 - y_2)*x_1 + (x_1*w**2 + x_2*w + x_3 - y_3)*(2*x_1*w +$$

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Solution: $(x_1*w + x_2 - y_2)*x_1 + (x_1*w**2 + x_2*w + x_3 - y_3)*(2*x_1*w +$
 $y_3)*(2*x_1*w + x_2)$

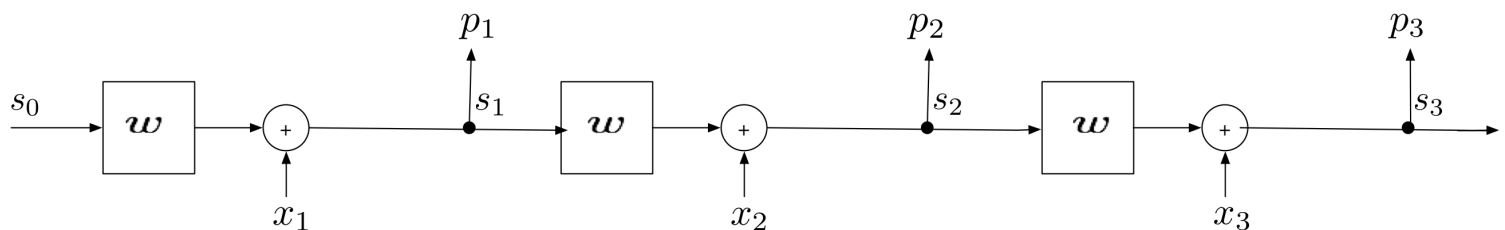
$$(x_1 \times w + x_2 - y_2) \times x_1 + (x_1 \times w^2 + x_2 \times w + x_3 - y_3) \times (2 \times x_1 \times w + x_2)$$

Explanation:

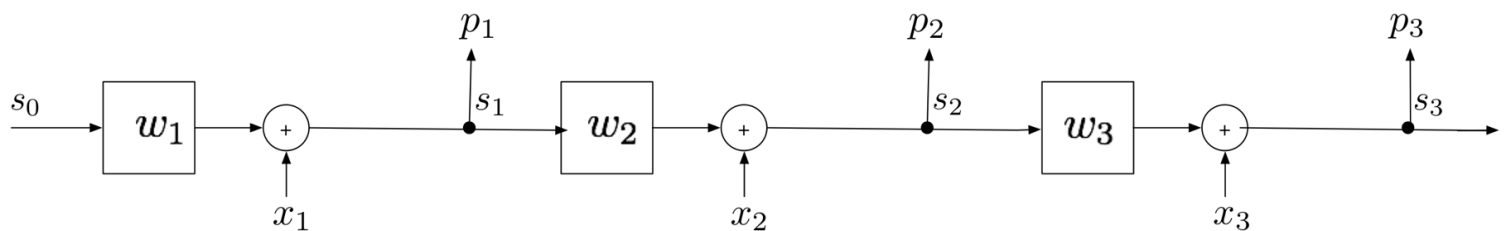
We can unroll the RNN equations to obtain $p_1 = x_1$, $p_2 = wx_1 + x_2$, and $p_3 = w^2x_1 + wx_2 + x_3$. Thus $\frac{\partial p_1}{\partial w} = 0$, $\frac{\partial p_2}{\partial w} = x_1$, and $\frac{\partial p_3}{\partial w} = 2wx_1 + x_2$. Finally, $\frac{dL_{\text{loss}}}{dw} = \sum_{t=1}^3 (p_t - y_t) \frac{\partial p_t}{\partial w}$, and we just need to plug in the expressions.

3) Weights

Here is the unrolled RNN from the previous problem. Let's call this the *single-w model*



Consider in addition a related model where instead of a single weight w , at each time step we can use a different weight w_t ; call this the *multi-w model*.



Assume $s_0 = 0$ throughout.

3A) Can the single-w model exactly represent a machine that takes $x = [1, 1, 1]$ and produces $y = [1, 2, 4]$? If so, what is the value of w ? If not, enter 'None'.

Enter a number or 'None' :

'None'

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Solution: 'None '

Explanation:

By plugging in the values for x_2 , x_3 , s_1 , s_2 , and s_3 for the equations for the second and third time steps, we can see that there is no single value of w that satisfies both equations.

3B) Can the multi- w model exactly represent a machine that takes $x = [1, 1, 1]$ and produces $y = [1, 2, 4]$? If so, what are the values of $[w_1, w_2, w_3]$ as a Python list? If not, enter 'None'.

Enter a list of 3 numbers or 'None' :

[0,-2,-6]

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Solution: ['anything', 1, 1.5]

Explanation:

We can plug in the values for s_t , s_{t-1} , and x_t for each timestep as we did for 2A and thus solve for w_t for each equation.

3C) Which of the following are reasons to prefer the single- w model (over the multi- w model)? Note when we say ``memorize a training sequence $[x_1, x_2, \dots, x_n]$ " we mean that the model can potentially produce any output sequence $[y_1, y_2, \dots, y_n] \in \mathbb{R}^n$ given this input.

Choose all that are true:

- ☐ It can represent a bigger class of sequences.
- ☒ It will generalize better with a small amount of data.
- ☒ It will capture the data's structure as a finite state machine.
- ☐ It can memorize any single training sequence

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3D) Which of the following are reasons to prefer the multi-w model (over the single-w model)?

Choose all that are true:

- ☒ It can represent a bigger class of sequences.
- ☐ It will generalize better with a small amount of data.
- ☐ It will capture the data's structure as a finite state machine.
- ☐ It can memorize any single training sequence

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